Sachin Deb

Ph.D. Student in Computer Science • Graduate Research Associate • Arizona State University

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Research Focus

Researcher building label-efficient, multimodal foundation-model methods for health: adapting pretrained models for medical imaging (3D CT), wearable and contactless physiological sensing, and clinical language to medical tasks with scarce labels, and testing that results hold for new people, new cohorts, and under-represented subgroups. Current work spans interpretable stress and craving detection in opioid use disorder, CT-based bladder cancer staging and report generation with Mayo Clinic, and zero-shot wearable sensing, backed by production ML industry experience.

Interests: medical foundation models • multimodal learning • wearable & contactless sensing • zero-shot learning • ML fairness

Education

Arizona State University 2025 – Present
Ph.D. in Computer Science | *GPA 4.00/4.00* | *Advisor: Prof. Asif Salekin* *Tempe, AZ*

Bangladesh University of Engineering and Technology (BUET) 2018 – 2024
B.Sc. in Computer Science and Engineering | *CGPA 3.64/4.00* *Dhaka, Bangladesh*

Manuscripts Under Review

- *Representation Is Not Enough: Body-Localized Thermal Evidence for Contactless Stress and Craving Sensing in Opioid Use Disorder.* Manuscript under review.
- *Reducing the Adaptation Gap Through Reachable Fisher Geometry.* Under review at ICLR 2027.

Research Experience

Graduate Research Associate, Ubiquitous & Intelligent Sensing (UIS) Lab 2025 – Present
Arizona State University | *Advisor: Prof. Asif Salekin* *Tempe, AZ*

FABLE-Therm: Contactless Stress and Craving Sensing from Thermal Video Under Review

Interpretable, label-efficient detection of stress and drug craving from a thermal camera, in a clinical population.

- Framed the task as weakly supervised evidence localization: from only a clip-level label, the model learns *which body regions and which moments* carry the signal, with no wearable-sensor supervision.
- Designed a multiple-instance learning architecture (temporal summaries, spatial self-attention, gated MIL pooling) that keeps evidence separated by region, time, and encoder, so each prediction traces back to its source.
- Fused several frozen thermal and vision foundation models through one shared aggregator: **0.938 AUROC** on held-out participants vs. **0.744** for separate per-encoder modules, with *fewer* trainable parameters; explained why in theory (separate branches learn incompatible coordinate systems).
- Embedding-level fusion beat feature- and prediction-level fusion of the same encoders (**0.938** vs. **0.736** / **0.735**).
- Built a cohort-transfer analysis that splits the drop on a held-out opioid-use-disorder cohort into representation mismatch (**0.158 AUROC** recovered with in-cohort data) and person-to-person variability (**0.169 AUROC** remaining), showing what more data can and cannot fix.
- Generalized to self-reported craving (**0.752 AUROC**) and validated on the public StressNet dataset.

Reachable Fisher Geometry for Fairer Parameter-Efficient Fine-Tuning ICLR 2027 Submission

Why fine-tuning large pretrained models helps some subgroups more than others, and a training fix that narrows the gap.

- Introduced the *reachable Fisher*: model curvature measured only along directions a LoRA or DoRA adapter can actually change, a principled lens on subgroup gaps after fine-tuning.
- Characterized the LoRA Jacobian kernel: guarantees for factorized adapters must be stated on the subspace that truly changes the model, not in redundant adapter coordinates (extended to DoRA).
- Proved when reachable-Fisher differences predict whether a subgroup gap grows or shrinks, showed the Fisher trace alone cannot guarantee this, and derived a matrix-free identity that makes auditing large models practical.
- Showed that the restricted operator norm of the reachable Fisher governs the worst-case (quadratic) change in subgroup gaps during fine-tuning.
- Trained and evaluated **306 models** across image and tabular tasks: higher reachable-Fisher trace matched the harder subgroup in **75.7%** of 1,218 evaluations, and a trace-matching training objective narrowed the best-to-worst subgroup gap in **all 30** task/encoder/adapter settings (**17.2%** median relative reduction).

Mayo Clinic: 3D CT Foundation Models for Bladder Cancer Staging ONGOING

Multimodal learning from volumetric CT and radiology reports, toward staging and structured report generation.

- Adapting and combining pretrained 3D CT foundation models (e.g., FMCIB, SwinUNETR, SegResNetDS) to separate muscle-invasive ($\geq$T2) from non-muscle-invasive bladder cancer from Mayo Clinic CT studies with limited labels.
- Learning from images and text together by pairing each CT volume with its radiology report through a medical text encoder, with pathology-confirmed stage as ground truth.
- Building one pipeline that localizes the bladder, segments the tumor, predicts stage, and drafts a structured radiology-style report, designed for small multi-planar CT urography cohorts and to scale with more data.

Clinical-Language and Wearable-Signal Alignment for Zero-Shot Detection ONGOING

Reading physiological signals through a clinical language model, for conditions where labeled data is rare.

- Aligning GatorTron (a clinical language model) with the signal encoder of NormWear (a wearable foundation model) so short electrodermal activity (EDA) windows are classified zero-shot for cannabis, alcohol, and craving.
- Developing domain-calibrated adaptation: a three-stage pipeline that specializes the generalist model to one physiological domain (e.g., craving, stress, epilepsy, cardiovascular) while keeping its pretrained knowledge.
- Evaluating with leave-one-dataset-out zero-shot tests across several wearable datasets against the frozen generalist.

Undergraduate Researcher

Bangladesh University of Engineering and Technology | *Advisor: Dr. Muhammad Ali Nayeem* *Dhaka, Bangladesh*

MOONLiHood: Federated Learning on Non-IID Client Data CODE

Privacy-preserving training across clients whose data cannot be pooled and is unevenly distributed.

- Combined model-contrastive learning, likelihood-based client weighting, and simulated annealing to decide how much each client should contribute when local data distributions differ.
- Reached about **71%** global test accuracy in CIFAR-10/100 experiments vs. about **61–65%** for FedAvg baselines.

Industry Experience

Chaldal Engineering (Y Combinator-backed) Jun 2024 – Jul 2025

Software Engineer, Level 2

- Built a personalized recommendation pipeline using a two-tower model, deployed with A/B testing, and improved multilingual semantic search in production.
- Engineered real-time (WebSocket) parts of a 10-minute delivery system, contributing to a **20% rise** in orders.
- Built and maintained catalog, search, and recommendation microservices on Kubernetes with CI/CD (Azure DevOps).
- Cut worst-case mobile navigation latency from **30 s to 5 s** on Android and iOS (state management, Hermes).

MYSOFT Limited Jan 2023 – Apr 2023

Engineering Intern

- Implemented authentication, sessions, catalog, cart, and order processing for a Spring Boot / Angular web platform.

Technical Skills

Health Data	Volumetric CT, radiology reports, wearable physiological signals (EDA), thermal video
Methods	Foundation-model adaptation, LoRA / DoRA, multimodal fusion, MIL, ViTs, segmentation, zero-shot
Evaluation	Held-out-participant and leave-one-dataset-out validation, subgroup auditing, large-scale ablations
ML Frameworks	PyTorch, Hugging Face (Transformers, PEFT), MONAI, timm, scikit-learn, TensorFlow / Keras
Languages	Python, C++, Java, C#, F#, JavaScript / TypeScript, Bash
Systems	Linux, CUDA, HPC / SLURM, Docker, Kubernetes, Azure DevOps, FastAPI, Git

Honors & Competitive Programming

- **ICPC:** 2nd place, 2023 Asia Dhaka Preliminary Contest; 7th place, 2022 Asia Dhaka Regional Contest.
- **Google Kick Start:** 310th, Round F 2022; 477th, Round A 2021.
- **Peak ratings:** Codeforces 2002 | LeetCode 2309 | CodeChef 2103.

Teaching, Mentoring & Service

- **Chief Judge**, Bangladesh Olympiad in Informatics, National Round (2025); **Problem Setter and Judge**, BUET Inter-University Programming Contests (2023, 2024).
- **Competitive Programming Trainer**, Bangladesh University of Business and Technology (2023–2025) and International University of Business Agriculture and Technology (2024–2025).